

Comprehensive Malaria Trends and Risk Analysis in Nigeria

Project Overview

This project focuses on analyzing malaria trends in Nigeria using healthcare datasets to generate meaningful public health insights and build an interactive Power BI dashboard.

The analysis explores:

- Malaria burden trends
 - Disease incidence patterns
 - Reported versus estimated malaria cases
 - Healthcare data quality challenges
 - Public health surveillance insights
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Project Objectives

The main objectives of this project are to:

1. Analyze malaria trends over time in Nigeria.
 2. Compare officially reported malaria cases with estimated malaria cases.
 3. Evaluate malaria incidence rates per 1000 population at risk.
 4. Assess healthcare data quality and reporting completeness.
 5. Build a professional healthcare analytics dashboard using Power BI.
 6. Generate public health insights from healthcare data.
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Datasets Used

Three healthcare datasets were used in this project.

1. Reported Malaria Cases Dataset

This dataset contains officially reported malaria cases and deaths.

Purpose

- Analyze reported malaria burden
- Study malaria mortality trends

- Evaluate healthcare reporting patterns
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2. Estimated Malaria Cases Dataset

This dataset contains estimated malaria case figures generated through health estimation models.

Purpose

- Compare reported and estimated malaria cases
 - Identify possible underreporting gaps
 - Evaluate surveillance system effectiveness
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3. Incidence per 1000 Population at Risk Dataset

This dataset contains malaria incidence rate information.

Purpose

- Measure malaria risk trends
 - Analyze disease burden relative to population risk
 - Study long-term malaria incidence patterns
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Tools Used

Microsoft Excel

Excel was used for:

- Data cleaning
 - Filtering Nigeria-specific records
 - Preliminary data inspection
 - Organizing healthcare datasets
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Microsoft Power BI

Power BI was used for:

- Data visualization
- Dashboard development
- KPI creation

- Healthcare data storytelling
 - Public health insight presentation
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Dashboard Components

The Power BI dashboard includes the following components.

1. Reported Malaria Cases Trend

Visual Type

Line Chart

X-Axis

Year

Y-Axis

Reported Malaria Cases

Purpose

To analyze trends in officially reported malaria cases over time.

2. Estimated Malaria Cases Trend

Visual Type

Line Chart

X-Axis

Year

Y-Axis

Estimated Malaria Cases

Purpose

To analyze estimated malaria burden trends over time.

3. Malaria Incidence Trend

Visual Type

Line Chart

X-Axis

Year

Y-Axis

Incidence Rate per 1000 Population at Risk

Purpose

To evaluate malaria risk trends within the population.

4. Reported vs Estimated Cases Comparison

Visual Type

Clustered Column Chart

X-Axis

Year

Y-Axis

- Reported Cases
- Estimated Cases

Purpose

To compare official reporting with estimated malaria burden and identify possible underreporting.

5. KPI Cards

The dashboard includes KPI cards for:

- Total Reported Cases
- Total Estimated Cases
- Average Incidence Rate

- Total Reported Deaths
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6. Data Quality Analysis

Visual Type

Pie Chart or Bar Chart

Variables

- Data Status
- Record Count

Purpose

To evaluate healthcare data completeness and reporting quality.

DAX Measures Used

Total Reported Cases

Total Reported Cases = SUM(reported_numbers[No_Of_Cases])

Total Estimated Cases

Total Estimated Cases = SUM(estimated_numbers[No_Of_Cases])

Average Incidence Rate

Average Incidence Rate = AVERAGE(incidence_per_1000_pop_at_risk[Incidence_Rate])

Total Reported Deaths

Total Reported Deaths = SUM(reported_numbers[No_Of_Deaths])

Key Healthcare Insights

The project investigates several important healthcare and public health themes.

Insight 1

Estimated malaria cases were consistently higher than officially reported cases, suggesting possible underreporting within malaria surveillance systems.

Insight 2

Malaria incidence remained persistently high across multiple years, indicating sustained public health burden.

Insight 3

Incomplete mortality records highlight potential healthcare data quality challenges.

Insight 4

Improved laboratory reporting systems could enhance malaria surveillance accuracy and intervention planning.

Laboratory Perspective

Accurate laboratory diagnosis and reporting are essential for reliable malaria surveillance, disease burden monitoring, and effective public health intervention planning.

This project demonstrates the connection between:

- Laboratory science
- Healthcare analytics

- Public health surveillance
 - Data-driven healthcare decision-making
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Skills Demonstrated

This project demonstrates:

- Healthcare data cleaning
 - Public health data analysis
 - Dashboard development
 - Healthcare storytelling
 - Trend analysis
 - Data interpretation
 - Research-oriented analytical thinking
 - Healthcare reporting analysis
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Career Relevance

This project supports future career opportunities in:

- Healthcare Data Science
 - Public Health Analytics
 - Epidemiology
 - Disease Surveillance
 - Healthcare Research
 - Healthcare AI and Analytics
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Repository Structure

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01-malaria-trends-nigeria/  
|  
├─ dataset/  
├─ powerbi-dashboard/  
├─ screenshots/  
├─ report/  
├─ presentation/  
└─ notes/
```

Conclusion

This project provides a comprehensive analysis of malaria trends and healthcare reporting patterns in Nigeria using healthcare datasets and Power BI visualizations.

The project highlights the importance of:

- Accurate disease reporting
- Public health surveillance
- Data quality assessment
- Laboratory diagnostics
- Healthcare analytics in disease monitoring and intervention planning

It also demonstrates how healthcare data science can support evidence-based public health decision-making.